

AI RESEARCH PAPER INTELLIGENT SYSTEM USING NATURAL LANGUAGE PROCESSING

1. Project Title

AI Research Paper Intelligent System Using Natural Language Processing

2. Abstract

The AI Research Paper Intelligent System is an NLP-based application developed to help users search, analyze, and understand research papers available on arXiv. The system retrieves research papers according to the topic entered by the user and performs various Natural Language Processing (NLP) tasks such as research paper summarization, keyword extraction, Named Entity Recognition (NER), category prediction, and similar research paper recommendation. This application reduces the time required to read lengthy research papers by providing the most important information in a structured format.

3. Introduction

Research papers are one of the most important sources of scientific knowledge. However, understanding multiple research papers requires considerable time and effort. Researchers and students often need a quick overview before deciding whether a paper is useful for their work.

This project applies Natural Language Processing techniques to automate the analysis of research papers. It searches research papers from the arXiv repository, extracts important information from the paper's abstract, generates a concise summary, identifies important keywords, recognizes named entities, predicts the research category, and recommends similar research papers. The system makes research paper analysis faster, easier, and more efficient.

4. Problem Statement

Reading and understanding research papers manually is a timeconsuming process. Users often spend significant time identifying relevant papers and extracting useful information. There is a need for an intelligent system that can automatically retrieve research papers, summarize their content, identify important information, and recommend related papers.

5. Objectives

- Retrieve research papers from the arXiv repository based on the user's searchtopic.
- Display research paper details such as title, authors, publication date, and PDFlink.
- Generate a concise summary of the research paper abstract.
- Extract important keywords using NLP techniques.
- Identify named entities such as organizations, dates, and persons using NamedEntity Recognition (NER).
- Predict the research category based on the paper content.
- Recommend similar research papers related to the selected paper.

6. Scope of the Project

The system is useful for students, researchers, faculty members, and professionals who regularly read research papers. It helps users quickly understand research papers without reading the complete document and supports efficient literature review.

7. Technologies Used

- Python
- Google Colab
- arXiv API
- spaCy
- NLTK
- Sumy
- YAKE

8. Libraries Used

- arxiv
- spacy
- nltk
- sumy • yake

9. System Architecture

User

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Enter Research Topic

↓

Search Papers using arXiv API

↓

Retrieve Research Papers

↓

User Selects a Research Paper

↓

Extract Research Paper Abstract

↓ Generate

Summary ↓

Extract Keywords

↓

Perform Named Entity Recognition (NER)

↓

Predict Research Category

↓

Recommend Similar Research Papers

↓

Display Final Output

10. Modules

Module 1: Research Paper Search

The user enters a research topic, and the system searches the arXiv repository to retrieve the most relevant research papers.

Module 2: Research Paper Details

The selected paper's title, authors, publication date, abstract, and PDF link are displayed.

Module 3: Research Paper Summarization

The abstract is summarized using extractive text summarization to provide the most important information in a shorter form.

Module 4: Keyword Extraction

Important keywords are extracted from the abstract using the YAKE keyword extraction algorithm.

Module 5: Named Entity Recognition (NER)

The system identifies named entities such as persons, organizations, locations, and dates using the spaCy NLP library.

Module 6: Category Prediction

The paper is classified into a research category using rule-based text analysis.

Module 7: Similar Research Paper Recommendation The system searches arXiv again using the selected paper's information and recommends similar research papers to the user.

11. Algorithm

Step 1: Start the program.

Step 2: Accept a research topic from the user.

Step 3: Search the arXiv repository for relevant research papers.

Step 4: Display the list of research papers.

Step 5: Allow the user to select a research paper.

Step 6: Retrieve the selected paper's abstract and details.

Step 7: Generate the research paper summary.

Step 8: Extract important keywords.

Step 9: Perform Named Entity Recognition (NER).

Step 10: Predict the research category.

Step 11: Recommend similar research papers.

Step 12: Display all information.

Step 13: End the program.

12. NLP Techniques Used

- Information Retrieval

- Text Processing
- Extractive Text Summarization
- Keyword Extraction
- Named Entity Recognition (NER)
- Rule-Based Text Classification

13. Working of the Project

The user enters a research topic. The system searches the arXiv database and retrieves relevant research papers. The user selects one paper from the list. The system extracts the paper's abstract and applies NLP techniques to generate a summary, identify keywords, recognize named entities, and predict the research category. Finally, the system recommends similar research papers and provides the PDF link for further reading.

14. Input

Research Topic entered by the user.

Example:

Artificial Intelligence

15. Output

The system displays:

- Research Paper Title
- Authors
- Publication Date
- Research Paper Abstract
- Generated Summary
- Extracted Keywords
- Named Entities
- Predicted Category
- Similar Research Papers
- PDF Link

16. Advantages

- Saves time while reading research papers.

- Retrieves the latest research papers from arXiv.
- Automatically generates summaries.
- Extracts important keywords.
- Identifies named entities using NLP.
- Recommends similar research papers.
- Easy to use and suitable for academic research.

17. Limitations

- The system currently works only with research papers available on arXiv.
- Category prediction is based on predefined rules.
- The summary is generated from the abstract instead of the complete research paper.
- Similar paper recommendations depend on the relevance of search results.

18. Future Scope

- Analyze complete research paper PDFs instead of only abstracts.
- Integrate transformer-based summarization models.
- Improve category prediction using machine learning techniques.
- Develop a web-based interface using Streamlit or Flask.
- Add multilingual support for research papers.
- Improve similarity recommendations using semantic embeddings.

19. Conclusion

The AI Research Paper Intelligent System demonstrates the practical application of Natural Language Processing in academic research. The system efficiently retrieves research papers, generates summaries, extracts keywords, performs Named Entity Recognition, predicts research categories, and recommends similar research papers. It reduces the effort required to analyze research papers and provides users with a faster and more organized way of understanding academic literature.

20. References

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